MIL-C-55481B(EL)
11 June 1969
Superseding
MIL-C-55481A(EL)
31 July 1967

MILITARY SPECIFICATION

CABLE ASSEMBLY, POWER ELECTRICAL CX-7453( )/U
CABLE ASSEMBLY, POWER ELECTRICAL CX-7705( )/U

1. SCOPE

1.1 Scope.- This specification covers Cable Assembly, Power, Electrical CX-7453( )/U and Cable Assembly, Power, Electrical CX-7705( )/U. (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-3530	Treatment, Mildew Resistant, for Thread and Twine.
MIL-M-13231	Marking of Electronic Items.
MIL-C-55442	Packaging and Packing of Cable Assemblies and Cord Assemblies.
MIL-C-55449	Connector, Plug, Electrical U-237( )/G

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal and Electronic Test Equipment.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.

FSC 5995

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DRAWINGS

ELECTRONICS COMMAND

SC-DL-115721 Cable Assembly CX-7705( )/U.
SC-DL-115724 Cable Assembly CX-7453( )/U (_FT_IN).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol shall be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Cable assemblies covered by this specification are as follows:

3.1.1 Cable Assembly CX-7453( )/U.- Cable Assembly CX-7453( )/U (_FT_IN) consists of a length of Cable WT-26( )/U with a Connector, Plug, Electrical U-237( )/G on each end.

3.1.2 Cable Assembly CX-7705( )/U.- Cable Assembly CX-7705( )/U consists of a 13 foot length of Cable WT-26( )/U with a Connector, Plug, Electrical U-237( )/G on one end and the other end is stripped and fanned.

3.2 Construction.- The cable assemblies shall be constructed in accordance with the requirements of this specification and the following:

(a) Drawing and Data List SC-DL-115721 for Cable Assembly CX-7705( )/U.

(b) Drawing and Data List SC-DL-115724 for Cable Assembly CX-7453( )/U.

3.3 Cleaning.- After assembly, the equipment shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material.

3.4 Marking.- Marking shall conform to MIL-M-13231. (See 4.3).

3.5 Connector.- Connector, Plug, Electrical U-237( )/G shall meet the requirements of MIL-C-55449(EL), except that the Pull, Moisture resistance and High humidity and high temperature tests shall be performed as part of this specification. (See 4.10).

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3.6 Electrical requirements.-

3.6.1 Continuity.- Electrical continuity shall exist through each conductor of the cable assembly. (See 4.6.1).

3.6.2 Dielectric strength.- There shall be no evidence of a voltage drop when tested in accordance with 4.6.2.

3.6.3 Insulation resistance.- When tested in accordance with 4.6.3, the insulation resistance shall be not less than 5,000 megohms except during the Moisture resistance test 4.8.1 and the High humidity and high temperature test 4.8.2, where it shall be not less than 1,000 megohms.

3.7 Mechanical requirements.-

3.7.1 Immersion.- After being subjected to the tests specified in 4.7.1, the assembly shall meet the requirements of Dielectric strength 3.6.2 and Insulation resistance 3.6.3, and there shall be no evidence of water in the connector housings.

3.7.2 Pull.- While being subjected to the test specified in 4.7.2, the assemblies shall show no straining of individual conductors or penetration of the grip through the cable jacket and shall meet the requirements of Continuity 3.6.1, Dielectric strength 3.6.2, and Insulation resistance 3.6.3. Upon completion of the Pull test, the assembly shall be subjected to the test specified in 4.7.1 and meet the requirements of Immersion 3.7.1.

3.8 Service conditions.- The equipment shall meet the following service conditions.

3.8.1 Moisture resistance.- When subjected to the test specified in 4.8.1, the assemblies shall meet the requirements of Insulation resistance 3.6.3, and there shall be no evidence of deterioration.

3.8.2 High humidity and high temperature.- When subjected to the test specified in 4.8.2, the assemblies shall meet the requirements of Insulation resistance 3.6.3, and there shall be no evidence of deterioration.

3.9 Treatment of textiles.- Prior to attachment of connectors, cut lengths of cable prepared for the connectors shall be immersed in a mildew inhibiting solution 3 inches beyond the edge at which the textile is exposed for a period of time necessary to obtain complete saturation, but shall not exceed 5 minutes. As soon as practical after immersion, the treated cable shall be dried so that it may be free of residual solvent. The solution shall be in accordance with MIL-T-3530, Type I, Class I.

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3.10 Workmanship.- The cable assemblies shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein: (See 4.9).

3.3 Cleaning
3.9 Treatment of textiles

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents. (See 4.3).

(b) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.4).

(2) Quality conformance inspection of preparation for delivery. (See 4.11).

4.3 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where required
Marking	3.4

4.4 Quality conformance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

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4.4.1 Group A inspection.- Each unit (100%) of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQLs indicated in table I. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table I - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual and mechanical	3.10	4.9	1.0%	4.0%
Electrical				
Continuity	3.6.1	4.6.1)	1.0% for	
Dielectric strength	3.6.2	4.6.2)	the group	
Insulation resistance	3.6.3	4.6.3)		

4.4.2 Group B inspection.- This inspection, including sampling, shall conform to table II and MIL-STD-105, Level S-3 for normal, tightened and reduced inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The AQL shall be as shown below.

Table II - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Immersion	3.7.1	4.7.1	1.5% defective

4.4.3 Group C inspection.- This inspection shall consist of the tests specified in table III, table IV and table V and shall be performed on samples that have been subjected to and met group A and group B inspection. Sample units shall be selected at random without regard to their quality except that the samples selected at the start of production shall be selected from the first units produced. Regardless of nomenclature or cable length, each item on the contract shall be considered a unit for purposes of the group C tests.

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4.4.3.1 Group C-1 inspection.- This inspection shall consist of the tests specified in table III and shall be performed on 4 units every month or every 1,000 units, whichever comes first.

Table III - Group C-1 inspection

Inspection	Req. Para.	Insp. Para.	Sampling
Pull	3.7.2	4.7.2	2 units mated

4.4.3.2 Group C-2 inspection.- This inspection shall consist of the tests specified in table IV and shall be performed every 3 months or every 3,000 units, whichever comes first.

Table IV - Group C-2 inspection

Inspection	Req. Para.	Insp. Para.	Sampling
High humidity and high temperature	3.8.2	4.8.2	4 units

4.4.3.3 Group C-3 inspection.- This inspection shall consist of the tests specified in table V and shall be performed once each year or every 12,000 units, whichever comes first.

Table V - Group C-3 inspection

Inspection	Req. Para.	Insp. Para.	Sampling
Moisture resistance	3.8.1	4.8.1	4 units

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4.4.3.4 Noncompliance.- The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.4.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.5 Preconditioning.- The contractor shall be permitted to precondition the equipment prior to performing any electrical tests except the electrical test performed during the Moisture resistance test 4.8.1 and High humidity and high temperature test 4.8.2. The preconditioning shall consist of removing surface moisture from the connector by wiping, drying or blowing. The application of heat shall be limited to 3 minutes or less. For those assemblies subjected to tests specified in Immersion 4.7.1, Moisture resistance 4.8.1 and High humidity and high temperature 4.8.2, unmated and without protective covers, a 24 hour waiting period in an ambient room temperature is permitted. However, if the contractor exercises this option of preconditioning, the contractor shall precondition all samples including those samples selected by Quality Assurance Representative immediately prior to testing.

4.6 Electrical tests.-

4.6.1 Continuity.- Electrical continuity of the cable assembly shall be verified by a point-to-point test to determine compliance with the requirements of 3.6.1.

4.6.2 Dielectric strength.- The assembly shall be tested in accordance with Method 301 of MIL-STD-202 to determine compliance with the requirements of 3.6.2. The potential, 3,500 volts rms (4,949 volts dc) shall be applied between each terminal in turn (except the ground terminal) and the remaining terminals connected together to the ground terminal and the housing ground.

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4.6.3 Insulation resistance.- The assembly shall be tested in accordance with Method 302, Condition B of MIL-STD-202 to determine compliance with the requirements of 3.6.3. The insulation resistance shall be measured between each terminal in turn (except the ground terminal) and the remaining terminals connected together to the ground terminal and the housing ground. The measurement shall be made after 1 minute electrification.

4.7 Mechanical tests.-

4.7.1 Immersion.- The assemblies shall be immersed to a depth of 3 feet of water for 16 hours to determine compliance with the requirements of 3.7.1. Fifty percent shall be mated, 25 percent shall be with protective cover in place and the remainder shall be without protective covers. (Minimum 4 connectors). (See 4.5).

4.7.2 Pull.- Two assemblies, with the connectors mated, shall be subjected to a static load of 425 pounds for a sufficient duration to permit electrical measurements. The load shall be applied between the mated connectors and a minimum of 2-1/2 feet of Cable WT-26( )/U on each connector. (See 3.7.2).

4.8 Service tests.-

4.8.1 Moisture resistance test.- Four assemblies, mated, unmated with protective cover in place and unmated without protective cover shall be subjected to 15 continuous 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. Insulation resistance measurements shall be made during the periods shown on the Standard, using Method 302, Condition A, of MIL-STD-202. The assemblies shall meet the requirements of 3.8.1 (See 4.5).

4.8.2 High humidity and high temperature.- Four assemblies, mated, unmated with protective cover in place and unmated without protective cover shall be placed in a chamber at 86°F ± 5° and a relative humidity of 92 to 98 percent. The temperature and humidity shall be maintained continuously for 5 days. While still at these conditions, the insulation resistance shall be measured using Method 302, Condition A, of MIL-STD-202. The assemblies shall meet the requirements of 3.8.2. (See 4.5).

4.9 Visual and mechanical inspection.- The assembly shall be examined for the defects listed in table VI. (See 3.10).

4.10 Connector.- Connector, Plug, Electrical U-237( )/G shall be tested prior to fabrication of cable assemblies to determine conformance with the requirements of 3.5.

4.11 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with MIL-C-55442(EL) to determine conformance to the requirements of section 5.

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Table VI - Classification of visual and mechanical defects

Classification	Defect
Major	Locking threads chipped or broken.
	Threads stripped.
	Sharp edges on metal parts.
	Coupler not operating freely.
	Metal parts chipped or broken.
	Contacts missing, broken, or bent.
	Contact assembly parts chipped or broken.
	Any foreign objects or material preventing the mating of connectors.
	Cuts or abrasions of rubber parts.
	Cut or ruptured cable.
	Marking incorrect.
	Scratches, cuts, abrasions, etc. with exposure of bare metal.
Minor	Screws missing.
	Finish not as specified.
	Scratches, cuts, abrasions etc., without exposure of bare metal.
	Finish adhesion-flaking, peeling or chipping.
	Wire grip missing.
	Scratches, cuts, abrasions etc., of contact assembly.
	Gasket not properly bonded to contact assembly.
	Abrasions on cable jacket.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with the requirements of MIL-C-55442(EL).

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6. NOTES

6.1 Intended use.- The cable assemblies are intended to supply electrical power to communication systems.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B or level C).

(d) Place of final inspection.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example; CX-7453W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

CUSTODIAN

ARMY EL

Preparing Activity

ARMY EL

Project 5995-A080

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